



Digital Maze 2D

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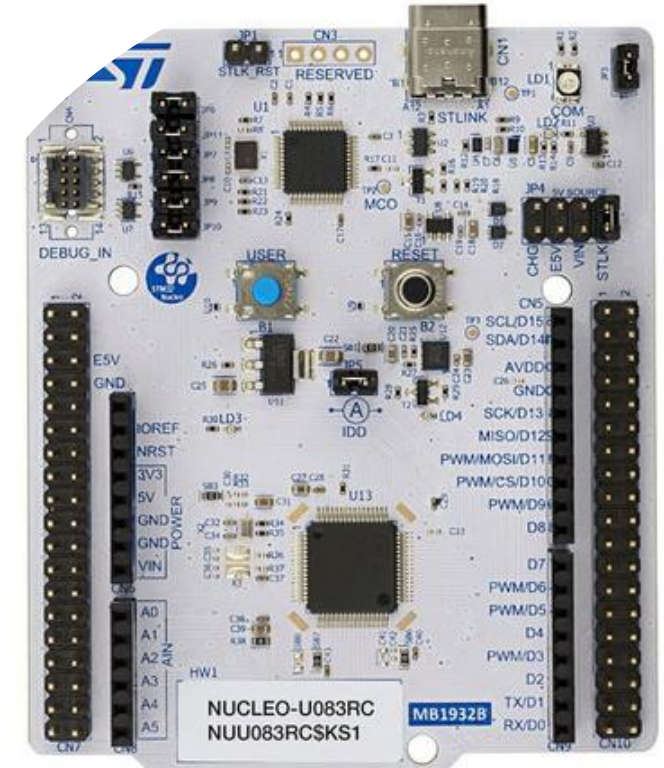
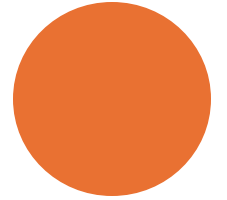
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Project Overview



Hardware Description (Components)

- **Microcontroller:** STM32L476RG (ARM Cortex-M4, 80MHz).
- **Display:** 1.8" TFT LCD with ST7735 Driver (128x128 pixels, 16-bit color).
- **Input:** 4 Directional Push-buttons (configured with internal Pull-up resistors).
- **Audio:** Piezoelectric Buzzer (for startup tones and victory music).
- **Power:** Powered via the Nucleo board's USB port (5V).



Pin Mapping (Hardware-Software Interface)



Peripheral	STM32 Pin	Software Role (HAL)
LCD SPI	PA5 (SCK), PA7 (SDA)	High-speed data transmission.
LCD Control	PA4 (CS), PB0 (DC), PB1 (RST)	Managing screen states and modes.
Buttons	PB3, PB4, PC8, PC9	Reading directional inputs (W, A, S, D).
Buzzer	PA10	Audio signal generation (Bit-banging).

Software Architecture

- **Language:** C (developed using STM32Cube HAL).
- **Core Modules:**
 - `main.c`: Main game loop, HUD updates, and victory checks.
 - `st7735.c`: Graphics driver (SPI command and data handling).
 - `maze.c` / `player.c`: Maze boundaries, collision detection, and coordinates.
- **Programming Techniques:**
 - **Non-blocking Timing:** Using `HAL_GetTick()` to manage time without freezing the CPU.
 - **Bit-banging:** Manually toggling GPIOs to generate sound frequencies.



Control Logic (HUD & Debouncing)

- **Software Debouncing:** Preventing multiple steps from a single press by implementing a 120ms timing window.
- **Heads-Up Display (HUD):**
- **Timer:** Updates every 1000ms using a non-blocking comparison.
- **Step Counter:** Increments only on successful, valid movements.
- **Victory Check:** Continuous comparison between player coordinates and goal coordinates within the `while(1)` loop.



Conclusions

- **Successful Integration:** Achieved stable high-speed SPI communication and responsive controls.
- **Performance:** Optimized use of STM32 processing power and memory (Flash/RAM).
- **Result:** A fully functional embedded system (Input → Processing → Output).
- **Future Scope:** * Implementation of DMA (Direct Memory Access) for faster graphics.
- Adding multiple levels and high-score saving via an SD card.

DEMO

